Lecture-2

Inverse Laplace transforms:

If the Laplace transform of a function $f(t)$ is $F(s)$ i.e., if $\mathcal{L}\{f(t)\} = F(s)$ then $f(t)$ is called the inverse Laplace transforms of $F(s)$ and we write

$$\mathcal{L}^{-1}\{F(s)\} = f(t).$$

Important formulae of Inverse Laplace transformation:

1	$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$	2	$\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}, n = 0,1,2, \ldots$
3	$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$	4	$\mathcal{L}^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$
5	$\mathcal{L}^{-1}\left\{\frac{a}{s^2-a^2}\right\} = \sinh at$	6	$\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$
7	$\mathcal{L}^{-1}\left\{\frac{a}{s^2+a^2}\right\} = \sin at$		

Some workout examples on Inverse Laplace transformation:

Example: 1	$\mathcal{L}^{-1}\left\{\frac{s^2+1}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s} + \frac{1}{s^3}\right\} = 1 + \frac{t^2}{2!} = 1 + \frac{t^2}{2}.$
Example: 2	$\mathcal{L}^{-1}\left\{\frac{1}{2s-5}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2(s-\frac{5}{2})}\right\} = \frac{1}{2}e^{\frac{5}{2}t}$
Example: 3	$\mathcal{L}^{-1}\left\{\frac{1}{s^2-16}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4}\frac{4}{s^2-4^2}\right\} = \frac{1}{4}\sinh 4t$
Example: 4	$\mathcal{L}^{-1}\left\{\frac{2s}{s^2-9}\right\} = 2\mathcal{L}^{-1}\left\{\frac{s}{s^2-3^2}\right\} = 2\cosh 3t$
Example: 5	$\mathcal{L}^{-1}\left\{\frac{4}{s-2} - \frac{s}{s^2-16} + \frac{4}{s^2-4}\right\}$ $= 4\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \mathcal{L}^{-1}\left\{\frac{s}{s^2-4^2}\right\} + 2\mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$ $= 4e^{2t} - \cosh 4t + 2\sinh 2t.$
Example: 6	$\mathcal{L}^{-1}\left\{\frac{5}{s} - \frac{3s}{s^2+16} + \frac{2}{s^2+4}\right\}$ $= 5\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 3\mathcal{L}^{-1}\left\{\frac{s}{s^2+16}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\}$ $= 5 - 3\cos 4t + \sin 2t.$

First translation property:

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$ **then** $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$.

Example: 01

$$\mathcal{L}^{-1}\left\{\frac{10}{(s+3)^4}\right\}$$
$$= 10\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^4}\right\}$$
$$= 10e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}$$
$$= 10e^{-3t}\frac{t^3}{3!} = \frac{10}{6}e^{-3t}t^3.$$

Example: 02

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+1}\right\}$$
$$= e^{2t}\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$
$$= e^{2t}\sin t.$$

Example: 03

$$\mathcal{L}^{-1}\left\{\frac{s-1}{(s-1)^2+4}\right\}$$
$$= e^{t}\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = e^{t}\cos 2t.$$

Example: 04

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2-25}\right\}$$
$$= e^{-2t}\mathcal{L}^{-1}\left\{\frac{s}{s^2-25}\right\}$$
$$= e^{-2t}\cosh 5t.$$

Example: 05

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^2-4}\right\}$$
$$= e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{s^2-2^2}\right\}$$
$$= e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{2}\frac{2}{s^2-2^2}\right\}$$
$$= \frac{1}{2}e^{-3t}\sinh 2t.$$

Example: 06

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+13}\right\} = \mathcal{L}^{-1}\left\{\frac{s+2-2}{(s+2)^2+3^2}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+3^2}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{(s+2)^2+3^2}\right\}$$
$$= e^{-2t}\,\mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\} - 2\,e^{-2t}\mathcal{L}^{-1}\left\{\frac{1}{s^2+3^2}\right\}$$
$$= e^{-2t}\,\mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\} - \frac{2}{3}e^{-2t}\mathcal{L}^{-1}\left\{\frac{3}{s^2+3^2}\right\}$$
$$= e^{-2t}\cos 3t - \frac{2}{3}e^{-2t}\sin 3t.$$

Example: 07

$$\mathcal{L}^{-1}\left\{\frac{2s+1}{s^2+4s+13}\right\} = \mathcal{L}^{-1}\left\{\frac{2(s+2)-3}{(s+2)^2+9}\right\} = \mathcal{L}^{-1}\left\{\frac{2(s+2)-3}{(s+2)^2+9}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{2(s+2)}{(s+2)^2+3^2} - \frac{3}{(s+2)^2+3^2}\right\} = 2e^{-2t}\cos 3t - 3\,e^{-2t}\sin 3t.$$

Example: 08

$$\mathcal{L}^{-1}\left\{\frac{s}{(s+3)^5} - \frac{2s+7}{s^2+4s+29}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{s+3-3}{(s+3)^5} - \frac{2(s+2)+3}{(s+2)^2+5^2}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{s+3}{(s+3)^5} - \frac{3}{(s+3)^5} - \frac{2(s+2)}{(s+2)^2+5^2} - \frac{3}{(s+2)^2+5^2}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{1}{(s+3)^4} - \frac{3}{(s+3)^5} - 2\frac{(s+2)}{(s+2)^2+5^2} - \frac{3}{5}\frac{5}{(s+2)^2+5^2}\right\}$$
$$= e^{-3t}\frac{t^3}{3!} - 3e^{-3t}\frac{t^4}{4!} - 2e^{-2t}\cos 5t - \frac{3}{5}e^{-2t}\sin 5t\,.$$

Inverse Laplace transformation using partial fraction:

Example: 01

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2-5s+6}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{1}{(s-3)(s-2)}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{1}{(s-3)}-\frac{1}{(s-2)}\right\}$$
$$=e^{3t}-e^{2t}.$$

Let, $\frac{1}{(s-3)(s-2)} \equiv \frac{A}{s-3}+\frac{B}{s-2}$

$\Rightarrow 1 = A(s-2)+B(s-3)$

$If\ s=2, B=-1$ and $if\ s=3, A=1$

Example: 02

$$\mathcal{L}^{-1}\left\{\frac{3s+1}{(s+1)(s^2+1)}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{3s+1}{(s+1)(s^2+1)}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{-1}{s+1}+\frac{s+2}{s^2+1}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{-1}{s+1}+\frac{s}{s^2+1}+\frac{2}{s^2+1}\right\}$$
$$=-e^{-t}+\cos t+2\sin t.$$

Let, $\frac{3s+1}{(s+1)(s^2+1)} \equiv \frac{A}{s+1}+\frac{Bs+C}{s^2+1}$

$\Rightarrow 3s+1 = A(s^2+1)+(Bs+C)(s+1)$

Comparing both sides, we get

$A+B=0, B+C=3\ and\ A+C=1$

By solving, we get

$A=-1, B=1$ and $C=2$

Example: 03

$$\mathcal{L}^{-1}\left\{\frac{4s+5}{(s-1)^2(s+2)}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{1/3}{s-1}+\frac{3}{(s-1)^2}-\frac{1/3}{s+2}\right\}$$
$$=\frac{1}{3}e^{t}+3t\,e^{t}-\frac{1}{3}e^{-2t}.$$

Let, $\frac{4s+5}{(s-1)^2(s+2)} \equiv \frac{A}{s-1}+\frac{B}{(s-1)^2}+\frac{C}{s+2}$

$\Rightarrow 4s+5 = A(s-1)(s+2)+B(s+2)$
$+C(s-1)^2$

Comparing both sides, we get

$A+C=0, A+B-2C=4$
$and\ -2A+2B+C=5$

By solving, we get

$A=\frac{1}{3}, B=3$ and $C=-\frac{1}{3}$.

Example: 04

$$\mathcal{L}^{-1}\left\{\frac{3s^2+13s+26}{s(s^2+4s+13)}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{2}{s}+\frac{s+5}{(s+2)^2+3^2}\right\}$$
$$=\mathcal{L}^{-1}\left\{\frac{2}{s}+\frac{s+2}{(s+2)^2+3^2}+\frac{3}{(s+2)^2+3^2}\right\}$$
$$=2+e^{-2t}\mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\}$$
$$+e^{-2t}\mathcal{L}^{-1}\left\{\frac{3}{s^2+3^2}\right\}$$
$$=2+e^{-2t}\cos 3t+e^{-2t}\sin 3t$$

Let, $\frac{3s^2+13s+26}{s(s^2+4s+13)} \equiv \frac{A}{s}+\frac{Bs+C}{s^2+4s+13}$

$\Rightarrow 3s^2+13s+26$
$= A(s^2+4s+13)+(Bs+C)s$

Comparing both sides, we get

$A+B=3, 4A+C=13\ and\ 13A=26$

By solving, we get

$A=2, B=1$ and $C=5$.

Inverse Laplace transformation using Cauchy residue theorem (CRT):

The formula for the Inverse Laplace Transform is

$$f(t)=\frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty}F(s)e^{st}ds=\sum_{\text{All poles of F(s)}}\operatorname*{Res}\left[F(s)e^{st}\right],$$

where $f(t)=0, t<0$.

If $s=a$ is a simple pole (where $F(s)$ is undefined)of $F(s)$, then

$$\operatorname*{Res}_{s=a}\left[F(s)e^{st}\right]=\lim_{s\to a}\left[(s-a)F(s)e^{st}\right],$$

and, if $s=a$ is a pole of order $n>1$, then

$$\operatorname*{Res}_{s=a}\left[F(s)e^{st}\right]=\lim_{s\to a}\frac{1}{(n-1)!}\frac{d^{n-1}}{ds^{n-1}}\left[(s-a)^n F(s)e^{st}\right].$$

Example 1: Find the inverse transform of $F(s)=\frac{1}{s+a}$ using CRT.

The function $\frac{e^{st}}{s+a}$ has a simple pole at $s=-a$. Hence

$$\operatorname*{Res}_{s=-a}\left[F(s)e^{st}\right]=\lim_{s\to -a}\left[(s+a)\frac{e^{st}}{s+a}\right]=e^{-at}.$$

and so $\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\}=\begin{cases}e^{-at}, & t\geq 0\\ 0, & t<0\end{cases}$.

Example 2: Find the inverse transform of $F(s)=\frac{1}{(s+a)^2}$ using CRT.

The function $\frac{e^{st}}{(s+a)^2}$ has a pole of order 2 at $s=-a$.

For the given problem, the required residue is

$$\frac{1}{(2-1)!}\lim_{s=-a}\frac{d}{ds}\left[(s+a)^2\frac{e^{st}}{(s+a)^2}\right]$$
$$=\lim_{s=-a}(te^{st})=te^{-at}.$$

And so $\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2}\right\}=\begin{cases}te^{-at}, & t\geq 0\\ 0, & t<0\end{cases}$.

Example 3: Find the inverse transform of $F(s)=\frac{1}{(s+a)^2(s+b)}$ using CRT.

The function $\frac{e^{st}}{(s+a)^2(s+b)}$ has a pole of order 2 at $s=-a$ and a simple pole at $s=-b$.

The residue at $s=-b$ is,

$$\operatorname*{Res}_{s=-b}\left[\frac{e^{st}}{(s+a)^2}\right]=\frac{e^{-bt}}{(a-b)^2}.$$

The residue at $s=-a$ is,

$$\frac{1}{(2-1)!}\lim_{s=-a}\frac{d}{ds}\left[\frac{e^{st}}{s+b}\right]=\lim_{s=-a}\left[\frac{te^{st}}{s+b}-\frac{e^{st}}{(s+b)^2}\right]$$
$$=\frac{te^{-at}}{b-a}-\frac{e^{-at}}{(b-a)^2}.$$

And so, $\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2(s+b)}\right\}=\frac{e^{-bt}}{(a-b)^2}+\frac{te^{-at}}{b-a}-\frac{e^{-at}}{(b-a)^2}, t\geq 0.$

MATLAB command to find inverse Laplace transformation:

$$F(s)=\frac{(s-5)}{s(s+2)^2}$$

>>syms t s

>> F=(s-5)/(s*(s+2)^2);

>>ilaplace(F, s, t)

ans =

(5*exp(-2*t))/4 + (7*t*exp(-2*t))/2 - 5/4

Exercise set: 2.1

Advanced Engineering Mathematics- Erwin Kreyszig(8th edition)

Page no:(211), Problem no: 29-32, 34, 35,37-40, 41-45

Find the inverse Laplace transformation of the following functions using Table or partial fraction or CRT or their combination. Also verify your results using MATLAB command:

	$F(s)$	$f(t)=\mathcal{L}^{-1}\{F(s)\}$
1	i) $\mathcal{L}^{-1}\left\{\frac{s^4+2s+1}{s^5}\right\}$; ii) $\mathcal{L}^{-1}\left\{\frac{1}{4s-3}\right\}$; iii) $\mathcal{L}^{-1}\left\{\frac{s-4}{s^2-8s+20}\right\}$; iv) $\mathcal{L}^{-1}\left\{\frac{1}{s^2-16}\right\}$; v) $\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\}$	
2	$\mathcal{L}^{-1}\left\{\frac{2-3s}{s^2+25}\right\}$	$\frac{2\sin 5t}{5}-3\cos 5t$
3	$\mathcal{L}^{-1}\left\{\frac{s-5}{s^2+10s+25}\right\}$	$e^{-5t}-10te^{-5t}$
4	$\mathcal{L}^{-1}\left\{\frac{4s-6}{s^2-6s+10}\right\}$	$4e^{3t}\left(\cos t+\frac{3\sin t}{2}\right)$
5	$\mathcal{L}^{-1}\left\{\frac{s}{s^2-5s+6}\right\}$	$3e^{3t}-2e^{2t}$
7	$\mathcal{L}^{-1}\left\{\frac{5s+2}{(s+1)(s-2)(s-3)}\right\}$	$-\frac{1}{4}e^{-t}-4e^{2t}+\frac{17}{4}e^{3t}$
8	$\mathcal{L}^{-1}\left\{\frac{s+7}{s(s+1)^2}\right\}$	$7-7e^{-t}-6te^{-t}$
9	$\mathcal{L}^{-1}\left\{\frac{2s-1}{s(s^2-4)}\right\}$	$\frac{1}{4}-\frac{5}{8}e^{-2t}+\frac{3}{8}e^{2t}$
10	$\mathcal{L}^{-1}\left\{\frac{s+1}{(s-1)(s^2+1)}\right\}$	$e^t-cost\ .$

Convolution theorem to find inverse Laplace transformation:

If $\mathcal{L}^{-1}\{F(s)\}=f(t)$ and $\mathcal{L}^{-1}\{G(s)\}=g(t)$ then

$$\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(u)g(t-u)du = \int_0^t f(t-u)g(u)du$$

Proof: See the page no. 56, Problem no. 21 (Schaum's Outline Series)

Example 1: Find inverse Laplace transformation of $F(s) = \dfrac{1}{s(s^2+1)}$ using convolution theorem.

Solution: $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}.\frac{1}{s^2+1}\right\}$

We have $\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1, \quad \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t,$

Then by convolution theorem,

$$\mathcal{L}^{-1}\left\{\frac{1}{s}.\frac{1}{s^2+1}\right\} = \int_0^t 1.\sin(t-u)du = \left[\frac{-\cos(t-u)}{-1}\right]_0^t$$
$$= \cos(t-t) - \cos(t-0) = 1 - \cos t.$$

Example 2: Find inverse Laplace transformation of $F(s) = \dfrac{1}{s^2(s+1)^2}$ using convolution theorem.

Solution: $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2}.\frac{1}{(s+1)^2}\right\}$

We have $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t, \quad \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = e^{-t}t$

Then by convolution theorem,

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)^2}\right\} = \int_0^t (t-u)ue^{-u}du \quad = \int_0^t (ut-u^2)e^{-u}du$$
$$= t\int_0^t ue^{-u}du - \int_0^t u^2e^{-u}du = te^{-t} + 2e^{-t} + t - 2.$$

Example 3: Find inverse Laplace transformation of $F(s) = \dfrac{1}{(s^2+4)(s-2)}$ using convolution theorem.

Solution: $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)(s-2)}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}.\frac{1}{s-2}\right\}$

We have $\mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} = \cos 2t\,, \quad \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} = e^{2t}$

Then by convolution theorem,

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)(s-2)}\right\} = \int_0^t \cos 2u\,.\,e^{2(t-u)}du$$

$$= \int_0^t \cos 2u\,.\,e^{2(t-u)}du = e^{2t}\int_0^t e^{-2u}\cos 2u\,du$$

$$= e^{2t}\left[\frac{e^{-2u}\left(-2\cos 2u + 2\sin 2u\right)}{(-2)^2+2^2}\right]_0^t; \quad \left[\text{Since, } \int e^{at}\cos bt\ dt = \frac{e^{at}(a\cos bt + b\sin bt)}{a^2+b^2}\right]$$

$$= e^{2t}\left[\frac{2e^{-2u}\left(\sin 2u - \cos 2u\right)}{8}\right]_0^t$$

$$= \frac{1}{4}e^{2t}\left[e^{-2t}\left(\sin 2t - \cos 2t\right) + 1\right]$$

$$= \frac{1}{4}(\sin 2t - \cos 2t) + \frac{1}{4}e^{2t}$$

$$\therefore\ \mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)(s-2)}\right\} = \frac{1}{4}(\sin 2t - \cos 2t) + \frac{1}{4}e^{2t}.$$

Example 4: Find inverse Laplace transformation of $F(s) = \frac{1}{(s+2)(s-3)^2}$ using convolution theorem.

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)(s-3)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+2}.\frac{1}{(s-3)^2}\right\}$$

We have $\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2t}$, $\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\} = te^{3t}$

Then by convolution theorem,

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)(s-3)^2}\right\} = \int_0^t e^{-2(t-u)}.e^{3u}.udu$$

$$= \int_0^t e^{-2t+5u}.udu$$

$$= \left[\frac{e^{-2t+5u}}{5}u - \frac{e^{-2t+5u}}{25}\right]_0^t$$

$$= \frac{1}{5}te^{-2t+5t} - \frac{1}{25}e^{-2t+5t} - \frac{1}{5}0.e^{-2t+0}2e^{-t} + \frac{1}{25}e^{-2t+0t}$$

$$= \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}.$$

Exercise set: 2.2

Find the inverse Laplace transformation of the following functions using convolution theorem. Also verify your results using MATLAB command:

	$F(s)$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$
1	$\mathcal{L}^{-1}\left\{\frac{2}{s(s^2+9)}\right\}$	$\frac{2}{9}(1-\cos 3t)$
2	$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)(s^2+1)}\right\}$	$\frac{1}{2}[\sin t - \cos t + e^{-t}]$
3	$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$	$\frac{t\sin at}{2a}$
4	$\mathcal{L}^{-1}\left\{\frac{1}{(s+4)(s-2)^2}\right\}$	$\frac{1}{6}te^{2t} - \frac{1}{36}e^{2t} + \frac{1}{36}e^{-4t}$

Inverse Laplace transformation associated with unit step function: (****)

Laplace transform of **unit step function**is $\mathcal{L}\{u(t-a)\} = \mathcal{L}\{u_a(t)\} = \frac{e^{-as}}{s}$

$$\mathcal{L}\{f(t).u(t-a)\} = e^{-as}\mathcal{L}\{f(t+a)\}$$

So, $\mathcal{L}^{-1}\left\{\frac{e^{-as}}{s}\right\} = u(t-a) = u_a(t)$.

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, then $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t) = f(t-a)u(t-a)$.

Some workout examples are given bellow:

Example 1:

Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s}\right\}$.

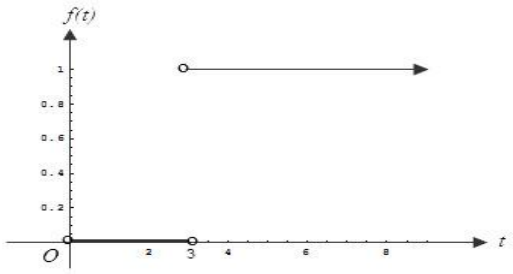
Solution: we know that

$$\mathcal{L}^{-1}\left\{\frac{e^{-as}}{s}\right\} = u(t-a) = u_a(t)$$

So,

$$f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s}\right\}$$

$$= u(t-3) = u_3(t) = \begin{cases} 0, & t < 3 \\ 1, & t > 3 \end{cases}.$$



Example 2:

Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2}\right\}$.

Solution:

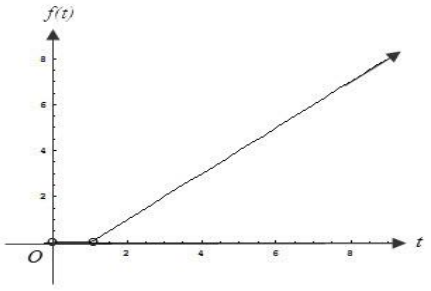
Let, $F(s) = \frac{1}{s^2}$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t = f(t)$.

We know that,

$\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t)$

So, $\mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2}\right\} = f(t-1)u_1(t) = (t-1)u_1(t)$

$$= \begin{cases} 0, & t < 1 \\ t-1, & t > 1 \end{cases}.$$



Example 3:

Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1}\right\}$.

Solution:

Let, $F(s) = \frac{1}{s^2+1}$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t = f(t)$.

We know that,

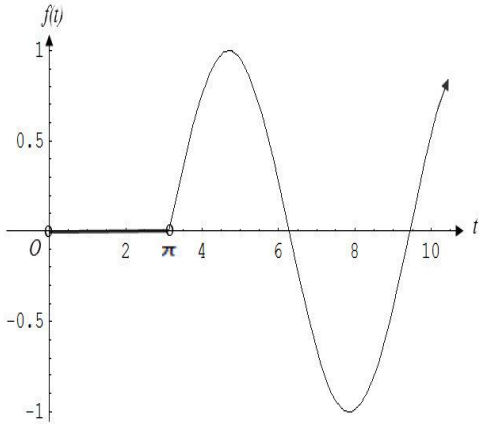
$\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t)$

So, $\mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1}\right\} = f(t-\pi)u_\pi(t)$

$$= \sin(t-\pi)\, u_\pi(t)$$

$$= \begin{cases} 0, & t < \pi \\ -\sin(\pi - t), & t > \pi \end{cases}$$

$$= \begin{cases} 0, & t < \pi \\ -\sin t, & t > \pi \end{cases}.$$



Example 4:

Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2+\pi^2} + \frac{e^{-2s}}{s^2+\pi^2} + \frac{e^{-4s}}{s^2}\right\}$.

Solution:

$\mathcal{L}^{-1}\left\{\frac{1}{s^2+\pi^2}\right\} = \frac{1}{\pi}\sin(\pi t)$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$.

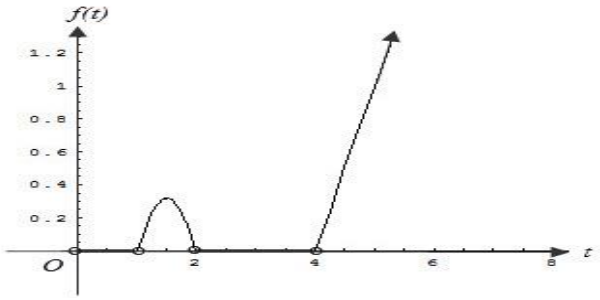
So, $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2+\pi^2} + \frac{e^{-2s}}{s^2+\pi^2} + \frac{e^{-4s}}{s^2}\right\}$

$$= \frac{1}{\pi}\sin\big(\pi(t-1)\big)u_1(t) + \frac{1}{\pi}\sin\big(\pi(t-2)\big)u_2(t) + (t-4)u_4(t).$$

Since, $\sin(\pi(t-1)) = -\sin(\pi t)$ and $\sin(\pi(t-2)) = \sin(\pi t)$, so the first two terms cancel each other when $t > 2$.

Hence, we obtain

$$f(t) = \begin{cases} 0, & 0 < t < 1 \\ -\frac{1}{\pi}\sin(\pi t), & 1 < t < 2 \\ 0, & 2 < t < 4 \\ t-4, & t > 4 \end{cases}.$$



Exercise set: 2.3

Find and sketch $f(t)$ where $f(t) = \mathcal{L}^{-1}\{F(s)\}$ are given bellow:

*** Ref: Advanced Engineering Mathematics- Erwin Kreyszig.

.

No.	$F(s)$	Answers: $f(t)$
1	$3\left(\frac{e^{-5s}}{s}\right)$	$3u_5(t) = \begin{cases} 0, & 0 < t < 5 \\ 3, & t \geq 5 \end{cases}$.
2	$4\left(\frac{e^{-3s}}{s^2}\right)$	$4(t-3)u_3(t) = \begin{cases} 0, & 0 < t < 3 \\ 4(t-3), & t \geq 3 \end{cases}$.
3	$\frac{e^{-6s}}{s-5}$	$e^{5(t-6)}u_6(t) = \begin{cases} 0; & 0 < t < 6 \\ e^{5(t-6)}; & t \geq 6 \end{cases}$